

INTERNSHIP REPORT

ON

DATA SCIENCE MASTER VIRTUAL INTERNSHIP

Submitted in partial fulfillment of the requirements for
BACHELOR OF TECHNOLOGY
Computer Science and Engineering
Specialization: Artificial Intelligence & Machine Learning

Submitted By
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Enrollment / Roll No.: 25scs1003005208

IILM UNIVERSITY, GREATER NOIDA

3rd Semester | Academic Session 2026

Internship Duration: June 2026 – August 2026

Mode: Virtual | Individual Internship

Supported by Siemens | AICTE–EduSkills

1. CANDIDATE'S DECLARATION

I, Sanskar Dubey, student of B.Tech Computer Science and Engineering with specialization in Artificial Intelligence & Machine Learning at IILM University, Greater Noida, hereby declare that this internship report entitled “Data Science Master Virtual Internship” is prepared on the basis of my learning and practical exposure during the eight-week virtual internship completed during June 2026 to August 2026.

I further declare that the report has been prepared for academic submission and that the work described in it represents my individual participation in the internship programme. The report summarizes the concepts, tools, activities, and learning outcomes covered during the internship, including data handling, data preparation, machine learning concepts, and practical exposure to Python, SQL, Excel, and RapidMiner.

I understand that any information presented as part of this report should be supported by the internship material, assignments, certificates, or other legitimate academic records available to me.

Place: Greater Noida

Date: _____

Signature of Candidate

Sanskar Dubey

Enrollment / Roll No.: 25scs1003005208

2. ACKNOWLEDGEMENT

I would like to express my sincere gratitude to IILM University, Greater Noida, for providing me with the opportunity to undertake this internship as a part of my academic learning. The internship provided me with an opportunity to connect classroom concepts with practical learning in the field of data science and machine learning.

I am thankful to AICTE and EduSkills for organizing the virtual internship programme and for providing a structured eight-week learning path. The internship covered applications and use cases, data engineering, machine learning, data preparation, and platform-related concepts. The programme also provided exposure to RapidMiner and data-oriented workflows.

I would also like to acknowledge Siemens for supporting the Data Science Master Virtual Internship. The learning structure helped me understand how data can be accessed, prepared, transformed, analyzed, and used in machine learning workflows.

Finally, I thank my teachers, mentors, and everyone who supported me during the internship. Their guidance and encouragement helped me complete the internship successfully and prepare this report.

Sanskar Dubey

3. INTERNSHIP COMPLETION CERTIFICATE

Certificate of Virtual Internship



Certificate issued for successful completion of the 8-week Data Science Master Virtual Internship.

TABLE OF CONTENTS

No.	Particulars	Page
1	Candidate's Declaration	i
2	Acknowledgement	ii
3	Internship Completion Certificate	iii
4	Project Description	1
4.1	Introduction	1
4.2	Organization Profile	2
4.3	Problem Statement	3
4.4	Project Objectives	3
4.5	Scope of the Project	4
4.6	Technologies and Tools Used	4
4.7	System Architecture	5
4.8	Methodology	6
4.9	Expected Outcomes	7
4.10	Certificates of Completion and Communication Proof	8
5	Bibliography / References	9

Note: Page numbers are arranged according to the report structure. They may shift slightly after final printing or if additional screenshots/college pages are inserted.

4. PROJECT DESCRIPTION

4.1 Introduction

The Data Science Master Virtual Internship was an eight-week AICTE–EduSkills programme supported by Siemens. The internship was completed virtually and individually during June 2026 to August 2026. It was designed to provide structured learning and practical exposure to data science, data engineering, machine learning, data preparation, and related platform concepts.

Data science combines data collection, preparation, analysis, visualization, statistical reasoning, and machine learning to support useful decisions. During the internship, the learning process followed a gradual progression from understanding applications and use cases to working with data, preparing datasets, developing machine learning understanding, and exploring deployment and platform administration concepts.

As a B.Tech student specializing in Artificial Intelligence & Machine Learning, the internship was relevant to my academic background. It helped strengthen my understanding of how data moves through a typical analytical workflow and how different tools can be used for practical data-oriented tasks.

Internship Overview

Particular	Details
Internship Title	Data Science Master Virtual Internship
Duration	8 Weeks
Period	June 2026 – August 2026
Mode	Virtual / Online
Participation	Individual
Supported By	Siemens
Programme	AICTE–EduSkills Virtual Internship
Academic Programme	B.Tech CSE – AI & ML

4.2 Organization Profile

EduSkills is an organization involved in industry-aligned skill development and internship programmes. The internship undertaken for this report was part of the AICTE–EduSkills virtual internship programme and was supported by Siemens. The programme was structured to provide students with exposure to practical and industry-relevant learning.

The offer letter for the internship describes the programme as an eight-week AICTE–EduSkills Virtual Internship. It states that the programme uses a week-wise learning structure, weekly assessments, project documentation or assignment submissions for selected modules, and a final assessment test. Successful completion of the required activities leads to a virtual internship certificate.

The internship was associated with the Siemens Data Science Master domain. The certificate provided after completion records the successful completion of the eight-week Data Science Master Virtual Internship during June–August 2026 and identifies Siemens as the supporting organization.

Week-wise Internship Structure

Week	Module	Learning Focus
1	Applications & Use Cases Professional	Identify business problems suitable for AI and machine learning.
2	Data Engineering Professional	Access, load, transform, and calculate data.
3	Machine Learning Professional	Build foundational machine learning skills.
4	Data Preparation Master	Advanced data handling and automation.
5	Machine Learning Master	Advanced machine learning techniques.
6	Applications & Use Cases Master	Deploy and monitor ML models.
7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
8	Platform Administration Master	Enterprise-level platform operations.

4.3 Problem Statement

Modern organizations generate large volumes of structured and unstructured data. The main challenge is not only collecting this data but also preparing it correctly, identifying useful patterns, and applying appropriate analytical or machine learning techniques. Students entering the field of artificial intelligence and machine learning therefore need practical exposure to the complete data science workflow.

The internship addressed this learning problem by providing a structured sequence of modules covering data engineering, data preparation, machine learning, applications and use cases, and platform administration. The practical learning also included exposure to tools such as Python, SQL, Excel, and RapidMiner.

4.4 Project Objectives

- Understand the role of data science in solving practical and business-oriented problems.
- Develop a basic understanding of data collection, loading, transformation, and preparation.
- Strengthen foundational knowledge of machine learning concepts and workflows.
- Gain practical exposure to Python for data-oriented tasks.
- Understand how SQL can be used to work with structured data.
- Use Excel for basic data organization, analysis, and presentation.
- Gain hands-on familiarity with RapidMiner and data science workflows.
- Understand the stages involved in deploying and monitoring machine learning models.
- Develop awareness of AI platform administration concepts through RapidMiner AI Hub modules.
- Complete the structured assessments and learning requirements of the eight-week internship.

4.5 Scope of the Project

The scope of this internship report is limited to the learning and practical exposure obtained during the eight-week Data Science Master Virtual Internship. It covers the major concepts and tools included in the internship curriculum rather than presenting the work as a production software system.

- Data handling and basic data engineering concepts.
- Data preparation, transformation, and organization.
- Foundational and advanced machine learning concepts introduced through the programme.
- Applications and use cases of AI and machine learning.
- Introduction to deployment and model monitoring concepts.
- Practical exposure to RapidMiner and RapidMiner AI Hub.
- Use of Python, SQL, and Excel in data-oriented learning activities.
- Individual completion of weekly learning and assessment activities.

4.6 Technologies and Tools Used

Python

Python was used as a programming language for data-oriented learning and for understanding how computational tasks can be implemented in a clear and structured manner. Its role in data science includes data manipulation, analysis, automation, and machine learning workflows.

SQL

SQL was used for understanding structured data and database-oriented operations. The internship learning helped connect database concepts with the broader data engineering and data analysis workflow.

Microsoft Excel

Excel was used for organizing and working with tabular data. It is useful for sorting, filtering, basic calculations, data inspection, and presenting information in a readable form.

RapidMiner

RapidMiner was an important part of the internship because the programme included Data Science Master modules and RapidMiner AI Hub administration. It provides a visual environment for building data science and machine learning workflows and helps learners understand the sequence of data preparation, modeling, evaluation, and related activities.

Data Structures

Data structures were part of the student's technical learning and support the broader understanding of how information can be organized and processed efficiently in programming and data science.

Summary of Tools

Tool / Technology	Purpose in Internship
Python	Programming and data-oriented tasks
SQL	Working with structured data and database concepts
Excel	Tabular data organization and basic analysis
RapidMiner	Visual data science and machine learning workflows
Data Structures	Organizing and processing data efficiently

4.7 System Architecture

The internship did not result in a single production application with a fixed software architecture. Instead, the practical learning can be represented through a general data science workflow. The following architecture summarizes the flow followed while understanding data-oriented and machine learning tasks.

Stage	Description
Data Sources	Structured datasets, tabular files, or other available data sources
Data Ingestion	Accessing and loading data for further processing
Data Preparation	Cleaning, transformation, formatting, and organization
Exploratory Analysis	Inspecting data and identifying useful patterns or relationships
Feature / Data Processing	Preparing suitable inputs for analytical or ML tasks
Machine Learning	Applying appropriate learning concepts or models
Evaluation	Checking model or workflow performance using suitable measures
Deployment / Monitoring	Understanding how models can be deployed and monitored
Result / Insight	Using the processed output for interpretation or decision support

In RapidMiner, several of these stages can be represented visually as connected operators in a workflow. This visual approach helped in understanding the sequence of operations without requiring every step to be implemented manually through code.

Architecture Flow

Data Source → Data Loading → Data Preparation → Analysis → ML Workflow → Evaluation → Deployment/Monitoring → Result

4.8 Methodology

The internship followed a structured week-wise methodology. Each module introduced a particular area of data science and machine learning, followed by learning activities and assessments. The overall methodology can be summarized into the following stages.

Step 1 – Understanding Applications and Use Cases

The internship began with understanding how artificial intelligence and machine learning can be used to solve practical problems. This stage focused on identifying suitable use cases and understanding the relationship between a problem and a possible AI/ML solution.

Step 2 – Data Engineering

The next stage focused on accessing, loading, transforming, and calculating data. This provided the foundation for understanding how raw data becomes usable for analysis.

Step 3 – Machine Learning Fundamentals

Foundational machine learning concepts were studied to understand how models learn patterns from data and how machine learning can be used for prediction or classification-oriented tasks.

Step 4 – Data Preparation

Data preparation was studied in greater depth. This included understanding the importance of clean, consistent, and properly formatted data before using it for analytical or machine learning tasks.

Step 5 – Advanced Machine Learning

The internship then progressed to advanced machine learning concepts, building on the fundamentals and improving understanding of model-building workflows.

Step 6 – Applications, Deployment and Monitoring

The programme introduced the process of deploying and monitoring machine learning models, connecting model development with practical usage.

Step 7 – RapidMiner AI Hub Administration

Platform administration concepts were introduced through RapidMiner AI Hub modules, providing an overview of how an AI platform can be configured and managed.

Step 8 – Enterprise Platform Operations and Final Assessment

The final stage covered enterprise-level platform operations followed by the final credential validation / internship grade point assessment.

4.9 Expected Outcomes

The internship was expected to improve both conceptual understanding and practical familiarity with data science tools and workflows. The major outcomes from the internship are summarized below.

- Improved understanding of the data science lifecycle.
- Better understanding of data engineering and data preparation activities.
- Stronger foundation in machine learning concepts.
- Practical familiarity with Python, SQL, and Excel for data-oriented tasks.
- Hands-on exposure to RapidMiner and visual machine learning workflows.
- Awareness of model deployment and monitoring concepts.
- Basic understanding of RapidMiner AI Hub administration.
- Improved ability to interpret data and organize analytical workflows.
- Completion of a structured eight-week industry-aligned learning programme.
- Greater confidence in continuing further study in Artificial Intelligence, Machine Learning, and Data Science.

Skills Developed

Skill Area	Learning Outcome
Data Handling	Understanding how data is accessed, loaded, transformed, and organized.
Data Preparation	Understanding the importance of clean and suitable data before analysis.
Machine Learning	Developing foundational and advanced conceptual understanding.
Programming	Using Python for data-oriented learning activities.
Database Concepts	Using SQL for structured data operations and understanding.
Spreadsheet Analysis	Using Excel for tabular data organization and basic analysis.
Visual Workflows	Using RapidMiner to understand data science workflow construction.
Platform Awareness	Understanding deployment, monitoring, and AI Hub administration concepts.

Internship Learning Reflection

The internship helped me understand that successful machine learning is not limited to selecting a model. Data quality, preparation, transformation, evaluation, and deployment are equally important. The structured sequence of the programme provided a broader view of the complete workflow and helped connect my academic studies in Artificial Intelligence and Machine Learning with practical data science concepts.

4.10 CERTIFICATES OF COMPLETION AND COMMUNICATION PROOF

The internship completion certificate is included in this report as documentary proof of successful completion. The certificate records the completion of the eight-week Data Science Master Virtual Internship during June–August 2026 at IILM University, Greater Noida, and identifies Siemens as the supporting organization.

Internship Offer Letter

The internship offer letter identifies the student as Sanskar Dubey, enrolled in B.Tech Computer Science and Engineering at IILM University, Greater Noida. It specifies the internship domain as Siemens Data Science Master and the duration as June 2026 – August 2026. It also states that the programme consists of a week-wise learning structure, weekly assessments, selected project/assignment submissions, and a final assessment.

Certificate Details

Particular	Recorded Information
Student Name	Sanskar Dubey
Institution	IILM University, Greater Noida
Internship	Data Science Master Virtual Internship
Duration	8 Weeks
Period	June – August 2026
Supporting Organization	Siemens

Additional screenshots or communication records may be inserted in this section if required by the university, such as internship portal progress, assignment completion, assessment results, or official communication received during the programme.

5. BIBLIOGRAPHY / REFERENCES

1. AICTE–EduSkills, Internship Offer Letter for Data Science Master Virtual Internship, June–August 2026.
2. AICTE–EduSkills, Certificate of Virtual Internship, Data Science Master Virtual Internship, 2026.
3. Siemens, Data Science Master learning programme materials and internship-supported learning resources.
4. RapidMiner, documentation and learning resources related to data science workflows and RapidMiner AI Hub.
5. Python Software Foundation, Python documentation and language reference.
6. Microsoft Excel documentation and learning resources for data organization and analysis.
7. SQL learning and reference materials used during the internship for structured data operations.

Appendix – Student Information

Field	Information
Name	Sanskar Dubey
Enrollment / Roll No.	25scs1003005208
University	IILM University, Greater Noida
Programme	B.Tech – Computer Science & Engineering
Specialization	Artificial Intelligence & Machine Learning
Semester	3rd Semester
Internship	Data Science Master Virtual Internship
Mode	Virtual
Participation	Individual
Duration	June 2026 – August 2026